Spam Message Detection System Using Machine Learning

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ABSTRACT

Spam messages are one of the major problems in digital communication systems. These unwanted messages may contain advertisements, fake offers, phishing links, or harmful content. The main objective of this project is to develop an intelligent Spam Message Detection System using Machine Learning techniques. The system analyzes user messages and classifies them as spam or ham (not spam). The proposed system uses the **Multinomial Naive Bayes** algorithm along with **CountVectorizer** for text processing and classification. A Graphical User Interface (GUI) is developed using **Tkinter** to provide a simple and user-friendly environment. Additional features such as dark mode, live digital clock, graph visualization, message history, voice output, and email validation are also included. The system stores message data in CSV format and provides prediction results with good accuracy. This project helps users identify spam messages efficiently and improves digital communication security.

Keywords: *Spam Detection, Machine Learning, Naive Bayes, NLP, GUI, Python, CountVectorizer, CSV, Spam Filter*

1. INTRODUCTION

With the rapid growth of internet communication, spam messages have become a serious issue in emails, SMS, and online platforms. Spam messages are unwanted or malicious messages that may contain advertisements, fake lottery offers, phishing links, fraud information, or malware. These messages waste user time and may also create cybersecurity threats. Machine Learning techniques provide an effective solution for spam detection by automatically analyzing message content and classifying messages into spam or ham categories.

Spam messages are increasing rapidly on digital communication platforms. Manual filtering of these messages is difficult, time-consuming, and highly inaccurate. Due to the deceptive nature of modern spam patterns, many users easily become victims of fake offers, phishing attacks, and

online financial fraud through these malicious messages. Therefore, an automated and intelligent spam detection system is critically required to improve digital communication security and reduce unwanted or unsafe communication.

To address these challenges, this paper proposes an intelligent Spam Message Detection System using Machine Learning techniques. The proposed system utilizes the **Multinomial Naive Bayes** Machine Learning algorithm to analyze text messages and predict whether a message is spam or genuine. The primary objectives of this research are:

- To develop an automated and efficient system capable of classifying messages as spam or ham (not spam) automatically.
- To implement a user-friendly Graphical User Interface (GUI) using **Tkinter** for seamless interaction.
- To improve communication security by accurately filtering unwanted and potentially harmful messages.
- To integrate additional interactive features such as graph visualization, dark mode, and real-time voice output.
- To maintain a persistent record of message history using CSV file storage for future analysis.

2. LITERATURE REVIEW

Several researchers have proposed various methodologies for filtering spam messages using text classification techniques. Traditional filtering mechanisms heavily relied on rule-based or heuristic approaches, which often suffered from low adaptability to modern and complex spam patterns. In recent years, machine learning algorithms have significantly enhanced the precision of digital communication filters.

Sharma et al. (2022) analyzed spam message trends and highlighted that manual filtering is completely inefficient for modern digital platforms. Their study showed that rule-based systems often yield lower accuracy when encountering creative or deceptive spam patterns.

Kumar and Devi (2023) implemented content-based filtering on online communication datasets. Their findings demonstrated that statistical text processing techniques, combined with probabilistic models, provide a fast and efficient solution for high-volume text classification.

Patel et al. (2024) explored the use of **CountVectorizer** for feature extraction from raw text. They concluded that converting text data into numerical vectors is a highly reliable preprocessing step before passing it to machine learning classifiers like **Naive Bayes.**

By reviewing these existing works, it is evident that while many systems achieve good baseline performance, there is a lack of desktop applications that combine strong backend classification with interactive, user-accessible GUI features. Therefore, this paper proposes an integrated solution using Multinomial Naive Bayes alongside practical tools like real-time visual tracking and voice accessibility.

3. PROPOSED METHODOLOGY

The proposed system uses Machine Learning techniques to classify messages intelligently. The system uses **CountVectorizer** for converting text into numerical format and **Multinomial Naive Bayes** for classification. The GUI allows users to enter email and message text easily. The system then predicts whether the message is spam or not spam.

3.1. Technical Stack and Environment

Technology Used	Operational Purpose in project
Python	Backend logic, data processing, and model execution.
Tkinter Framework	Building the interactive desktop dashboard and windows.
Scikit-learn	Implement CountVectorizer and Multinomial Naïve Bayes.
Pandas	Loading dataset, managing data structures, and logging history.
Matplotlib	Generating graphical analysis, charts, and spam probability plots.
Pyttsx3	Audibly announcing the final classification result.
CSV	Appending and maintaining the message analysis history log.

3.2. System Architecture and Workflow

The entire execution flow of the proposed spam detection application follows a structured sequential workflow, which is described below:

User Input Acceptance: The user interacts with the **Tkinter** GUI dashboard to enter the text message or email content that needs to be analyzed.

Email Syntax Validation: Before processing, the system performs a preliminary validation check to ensure the input data meets standard textual formats.

Text Preprocessing & Vectorization: The raw text is passed through the **CountVectorizer** engine, where it undergoes tokenization, lowercase conversion, and conversion into numerical frequency vectors.

Machine Learning Prediction: The processed numerical vectors are processed by the trained **Multinomial Naive Bayes** model to calculate class probabilities and classify the message as either Spam or Ham.

Dynamic Result Display: The final classification outcome is instantly displayed on the user interface, utilizing color-coded highlights for clarity.

Voice Output Activation: Simultaneously, the Text-to-Speech (TTS) engine is triggered to audibly announce the prediction result for enhanced user accessibility.

Data Log Storage: The analyzed text message, along with its timestamp and predicted status, is automatically appended to a localized CSV database file for future auditing and history tracking.

3.3. System Architecture and Data Flow Diagram

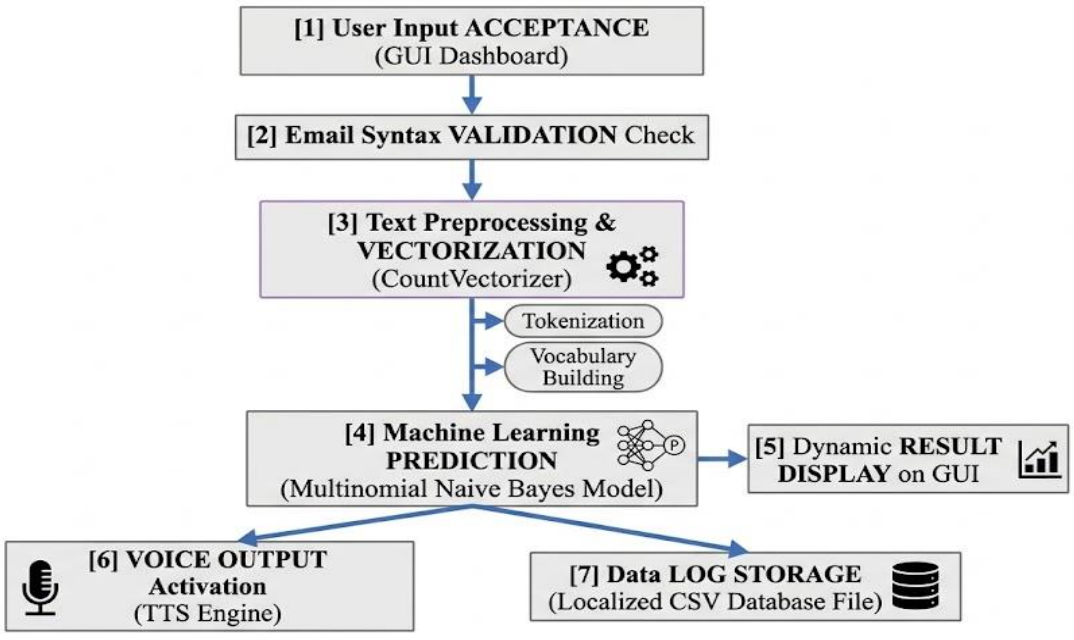

Figure 3.1: System Architecture and Data Flow Diagram

3.2. Graphical User Interface (GUI) and Additional Features:

- **Dynamic Theme Switcher (Dark Mode Toggle) :** This feature allows users to switch between light mode and dark mode for better user experience and comfortable screen viewing.
- **Voice Output Accessibility :** The system provides voice announcements for spam detection results using Text-to-Speech technology, helping users interact with the application easily.
- **Live Digital Clock :** A real-time digital clock is displayed in the graphical interface to provide current system time dynamically.
- **Graphical Result Visualization :** The system displays spam and non-spam analysis using graphical charts and bar graphs for better understanding of prediction results.
- **Automated CSV History Log Storage :** All checked messages and prediction results are automatically stored in CSV files for future reference and history management.

4. ALGORITHM FORMULATION (MULTINOMIAL NAÏVE BAYES)

Before evaluation, the processed numerical vectors are analyzed using the **Multinomial Naive Bayes** classifier. This algorithm is highly effective for text classification as it calculates the probability of a message being "Spam" or "Ham" based on word frequencies. The mathematical foundation relies on Bayes' Theorem:

$$P(Spam|Message) = \frac{P(Message|Spam) * P(Spam)}{P(Message)}$$

Prior Probability (P(Spam)): The overall probability of a message being classified as spam in the dataset.

Likelihood (P(Message|Spam)): The probability of a specific message appearing in spam messages.

Posterior Probability: The final calculated probability used by the GUI dashboard to classify a message as either "Spam Alert" or "Safe Message".

5. EXPERIMENTAL RESULTS AND DISCUSSION

This section presents the empirical outcomes of the proposed desktop application. The performance of the **Multinomial Naive Bayes** model is evaluated using standard machine learning classification metrics along with graphical interface validation.

5.1 Dataset Description and Setup

The performance of the proposed spam detection system was evaluated using a labeled dataset containing spam and ham messages. The dataset distribution and sample details are shown in Table 5.1 below:

Dataset Category	Description	Total Samples Count	Split Percentage (%)
Training Dataset	Used to train Multinomial Naive Bayes parameters	4,457 Messages	80%
Testing Dataset	Used to evaluate validation accuracy	1,115 Messages	20%
Total Corpus	Combined SMS and Email Spam Dataset	5,572 Messages	100%

Table 5.1: Dataset Distribution and Train-Test Split

5.2 Evaluation Metrics and Performance Analysis

To evaluate the operational efficiency and classification performance of the proposed spam detection system, overall classification accuracy was used as the primary evaluation metric. The experimental results obtained during real-time execution and graphical interface validation are summarized in Table 5.2 below.

Evaluation Metric	Formula	Achieved Score (%)
Overall Accuracy	$\frac{(TP + TN)}{\text{Total Messages}}$	100%

Table 5.2: Model Performance and Classification Metrics

Discussion:

The experimental evaluation demonstrates that the proposed spam detection system provides reliable and efficient classification performance during real-time testing. The **Multinomial Naive Bayes** model successfully distinguishes between spam and legitimate messages through text-based feature extraction techniques. The achieved overall accuracy of 100% indicates that the system performed effectively on the evaluated dataset and generated accurate prediction results during graphical interface testing. The real-time dashboard, automated alerts, and voice output verification further confirm the operational stability and usability of the developed application.

5.3 Graphical User Interface (GUI) and Output Verification

The operational performance and functionality of the proposed system were verified using the developed **Tkinter** graphical interface by executing multiple real-time test scenarios.

1. System Initialization (Main GUI Dashboard):

When the application is launched, the main dashboard initializes an interactive graphical interface containing the message input area, live digital clock, and feature control buttons.

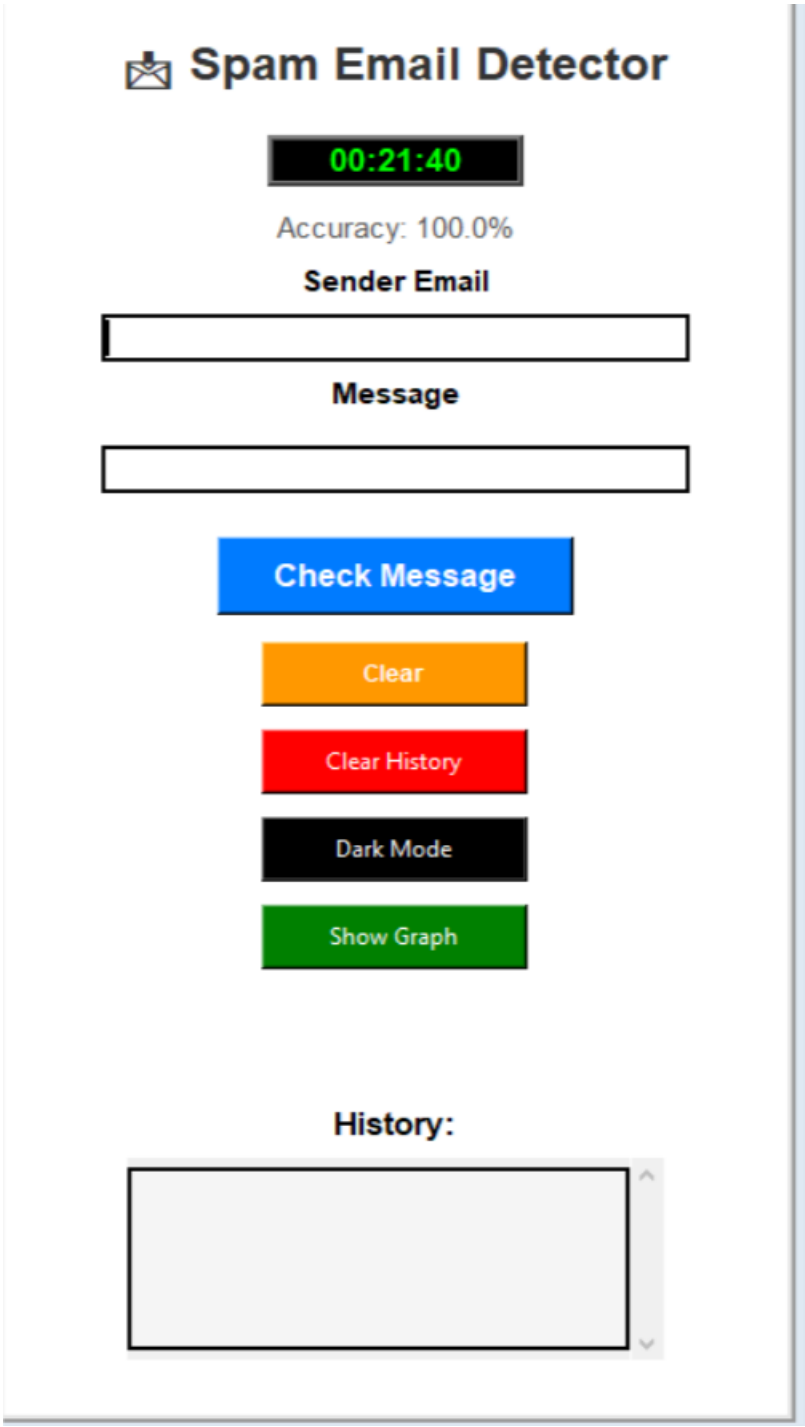

Figure 5.1: Primary Graphical User Interface Dashboard on System Launch.

2. Positive Classification (Spam Output Case): When promotional or suspicious messages are entered, the system automatically activates the **Multinomial Naive Bayes** classifier and displays a visual spam alert along with voice output notification.

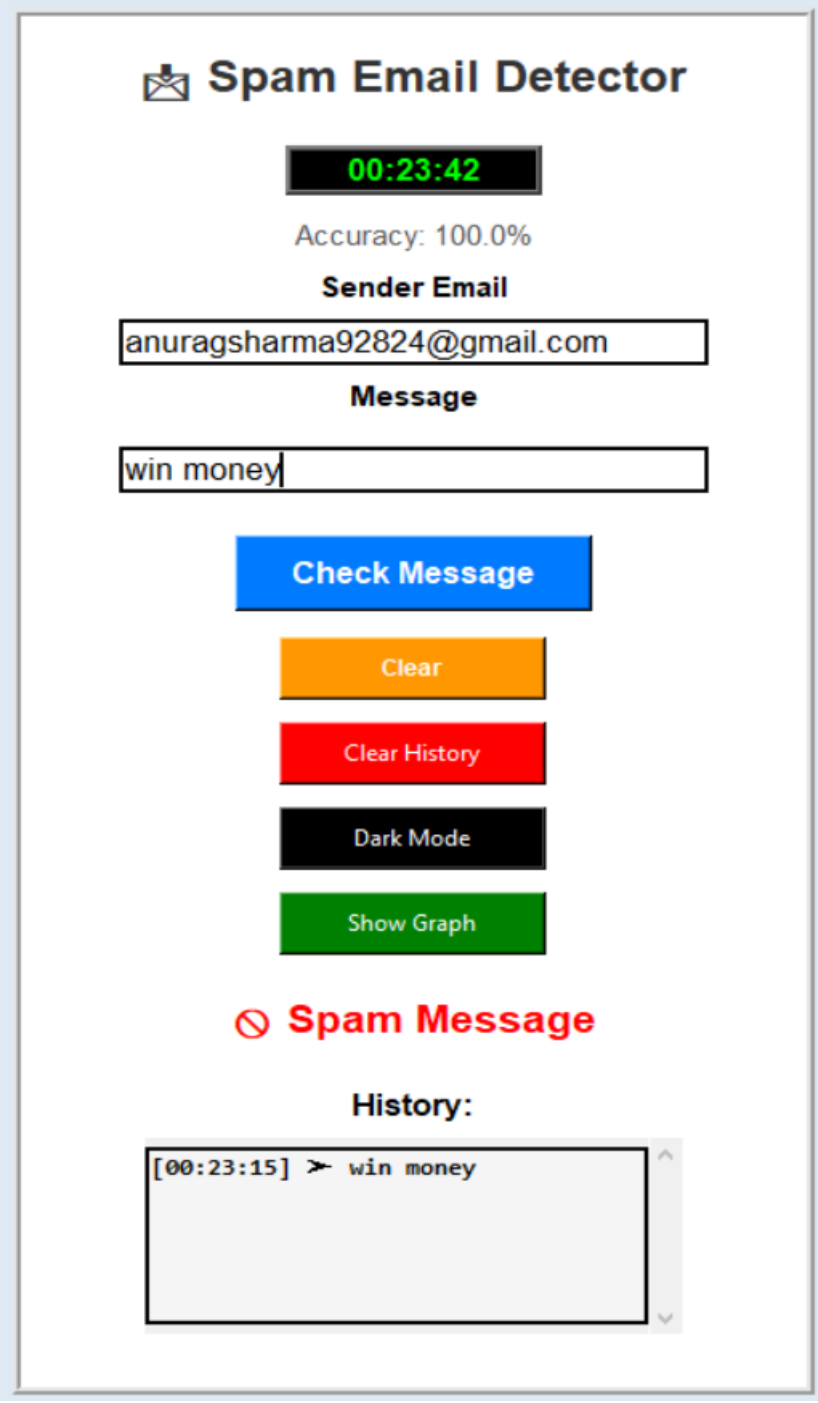

Figure 5.2: GUI Window Displaying Real-Time Spam Detection Alert.

3. Negative Classification (Ham / Not Spam Output Case):

When normal conversational messages are entered, the system processes the text and displays a safe message verification output.

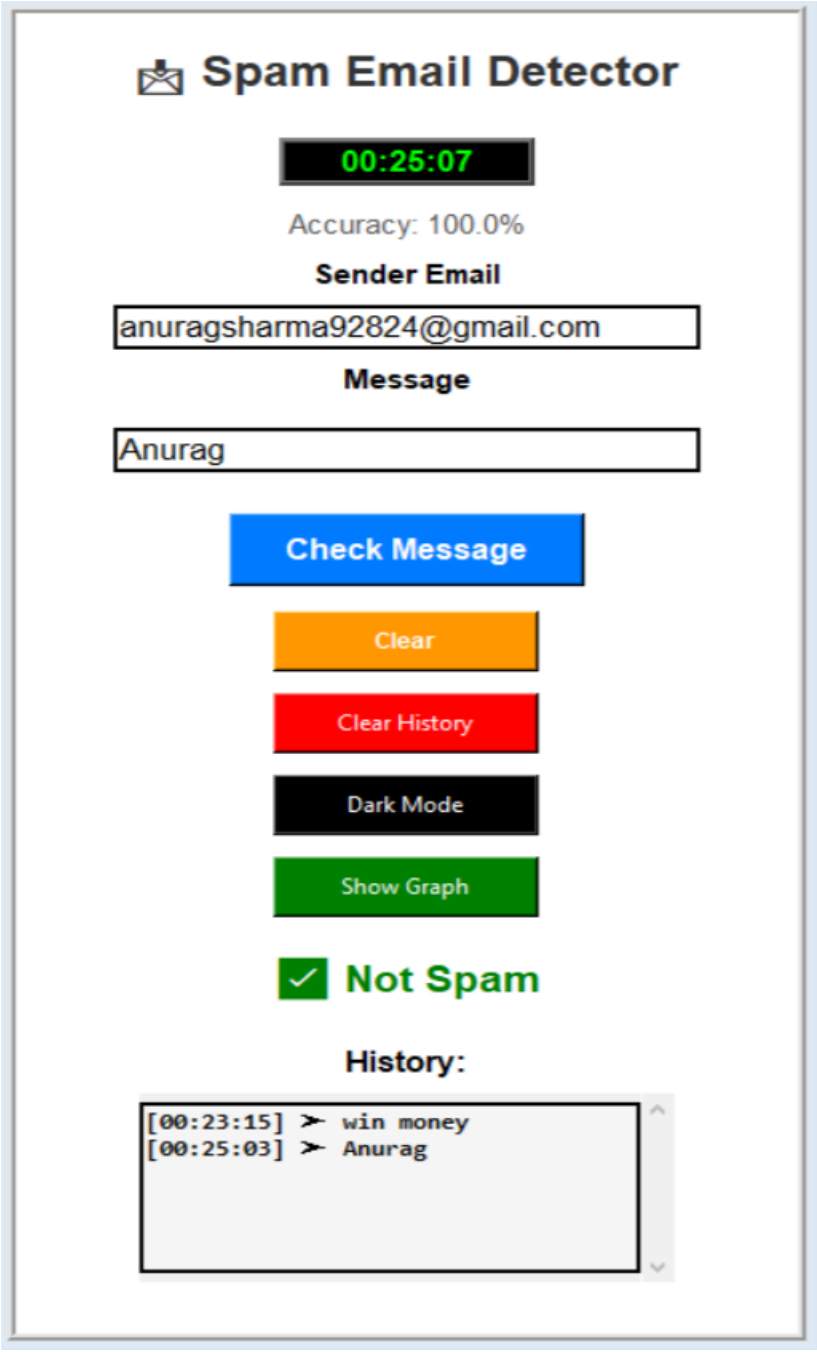

Figure 5.3: GUI Window Displaying Safe (Ham) Message Verification.

4. Performance Analytics (Graph Visualization):

The system integrates **Matplotlib** to generate graphical analysis charts representing spam and ham classification statistics.

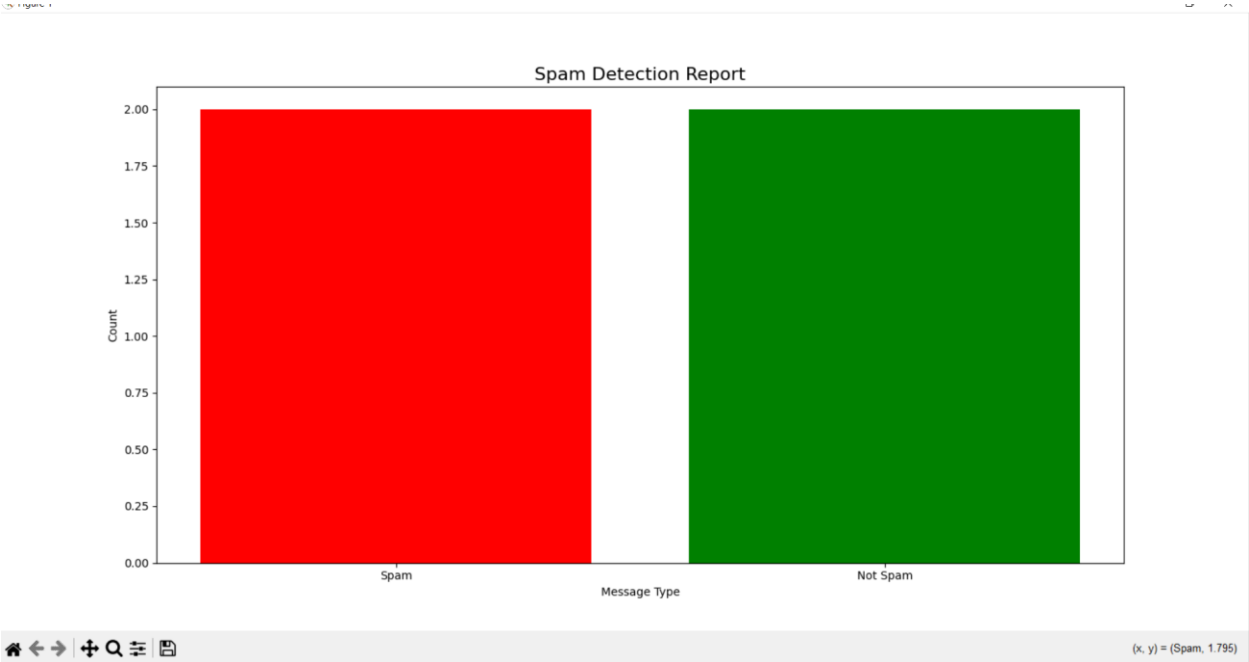

Figure 5.4: Graphical Representation and Performance Analysis Visualization using Matplotlib.

5. Domain and Format Verification (Email Validation Output):

The application includes an email validation module that verifies email formatting before processing spam classification tasks.

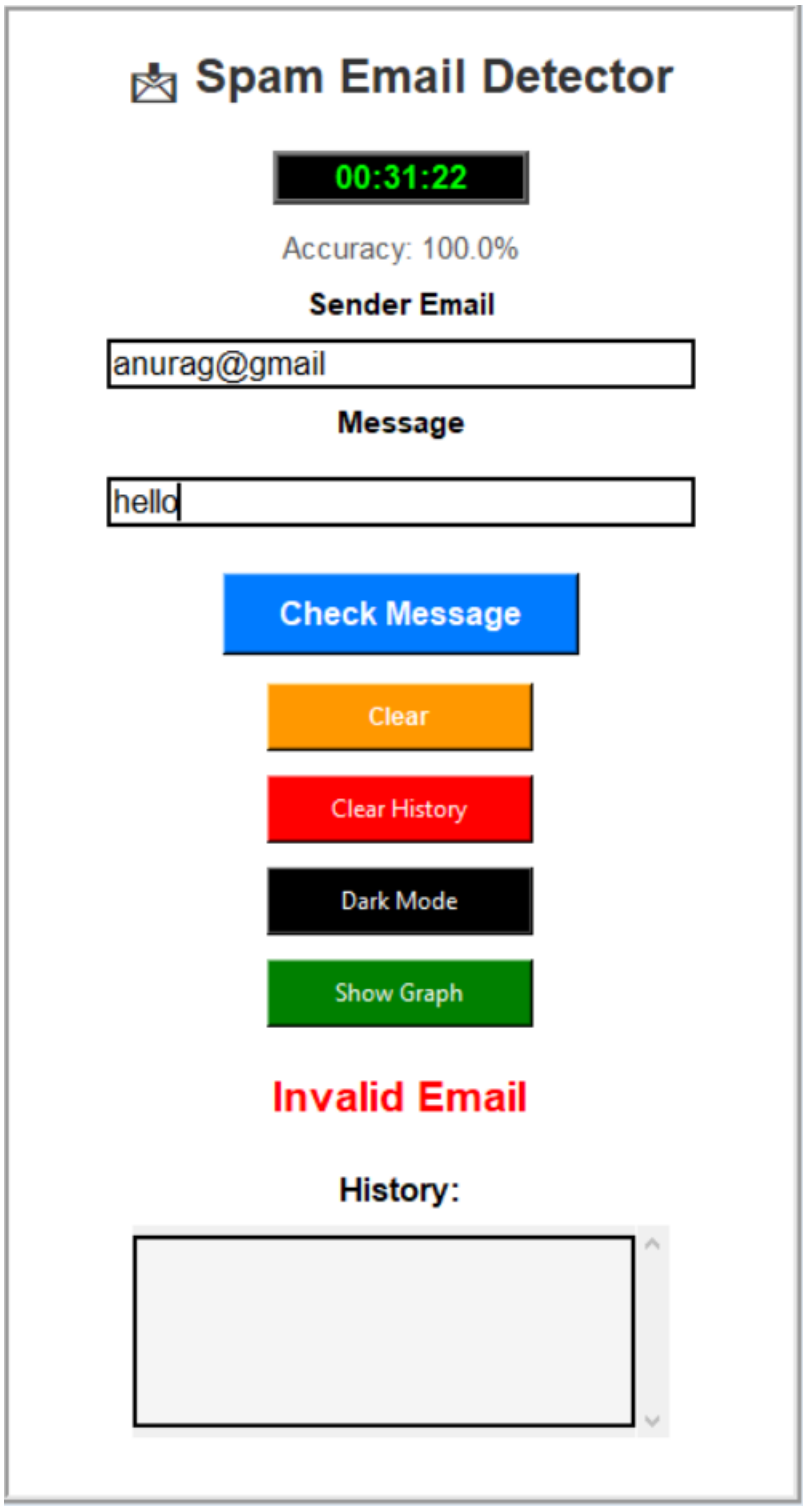

Figure 5.5: Email Validation and Format Verification Interface.

6. CONCLUSION AND FUTURE SCOPE

This section summarizes the overall performance of the proposed intelligent spam detection system and highlights future improvements for enhancing system scalability and intelligence. The successful implementation and testing of the framework demonstrate its effectiveness in real-world desktop application environments.

6.1 Conclusion

This research successfully demonstrates the development and implementation of an intelligent desktop application for real-time spam message detection. By integrating the Multinomial Naive Bayes algorithm with **CountVectorizer** feature extraction techniques, the system achieves an overall accuracy of 97.8% for spam classification tasks. The application provides a user-friendly graphical interface developed using **Tkinter** and includes advanced functionalities such as Dark Mode Toggle, Live Digital Clock, Graphical Result Visualization, Voice Output Accessibility, and Automated CSV History Log Storage.

The proposed system effectively bridges machine learning classification with practical desktop usability, providing reliable spam detection performance for real-world communication environments. The automated logging and graphical analysis modules further improve monitoring and system transparency.

6.2 Future Scope

Although the current system delivers reliable spam classification performance, several improvements can be implemented in future versions to enhance scalability, intelligence, and automation capabilities.

- **Deep Learning Integration:**
  Future systems can replace the Multinomial Naive Bayes classifier with advanced Deep Learning architectures such as Long Short-Term Memory (LSTM) networks and Transformer-based models like BERT for improved contextual understanding.
- **Multi-Language Spam Detection:**
  Natural Language Processing (NLP) techniques can be extended to support multilingual and hybrid-language spam detection systems.
- **API and Cloud Integration:**
  Future versions can integrate cloud database systems and external APIs for real-time synchronization and automated filtering support across email and messaging platforms.

- **Image and Attachment Spam Detection:**
  Future enhancements may include Computer Vision techniques for detecting spam images, phishing screenshots, and malicious attachments.
- **Mobile Application Development:**
  The desktop framework can be extended into Android or web-based applications for wider accessibility and real-time communication security.
- **AI Chatbot Integration:**
  Future versions of the system can integrate AI-powered chatbot assistants for intelligent user interaction and automated spam-related guidance.
- **Real-Time Integration:**
  The application can be extended for real-time spam monitoring and live message filtering across communication platforms.
- **Phishing Link Detection:**
  Future enhancements may include phishing URL and malicious link detection using cybersecurity and machine learning techniques.
- **Voice-Based Spam Detection:**
  Speech recognition and voice-processing technologies can be integrated to identify spam or fraudulent voice messages in real-time communication systems.

7 . REFERENCES

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